AI Mitigations Repository

Frameworks Slide Deck

Slide Deck

MIT AI Risk Index

About the AI Mitigations Repository

The Repository is a **database** and **a taxonomy** of AI risk mitigations

We compiled the database through a **systematic search** for existing frameworks, taxonomies, and other classifications of AI risks.

This slide deck presents the frameworks from the **13 included documents**.

For more information:

- 📃 [Read the research report]()
- 🌐 [Visit the website]()
- 📊 [Explore the repository]()

About the Frameworks

The frameworks of AI risk mitigations aim to **synthesize knowledge** on AI risks across academia and industry, and **identify common themes and gaps in our understanding** of AI risk mitigations.

This slide deck provides a **holistic view** of how AI risk mitigations are currently conceptualised. Readers can use it to **understand the variety of ways in which mitigations have been categorised** by various authors, and **bookmark particularly relevant frameworks** for future use.

We selected the documents in this deck based on:

- Their focus on presenting a structured taxonomy or classification of AI risk mitigations.
- Their coverage of mitigations across industry sectors.
- Their proposition of an original framework.

Table of Contents

Towards Best Practices in AGI Safety and Governance: A Survey of Expert Opinion

1. Survey of 92 AI safety experts found 85% average agreement on 50 safety practices for advanced AI development
2. 98% consensus on core measures: risk assessments, capabilities evaluations, third-party audits, safety restrictions, and red teaming
3. AGI lab experts agreed more than academics/civil society, but no significant disagreements emerged on specific practices
4. Results help AGI labs identify safety gaps and encourage commitments to audits and risk management
5. Findings inform US/EU/UK regulations and international standards, applicable beyond AGI labs to general-purpose AI developers

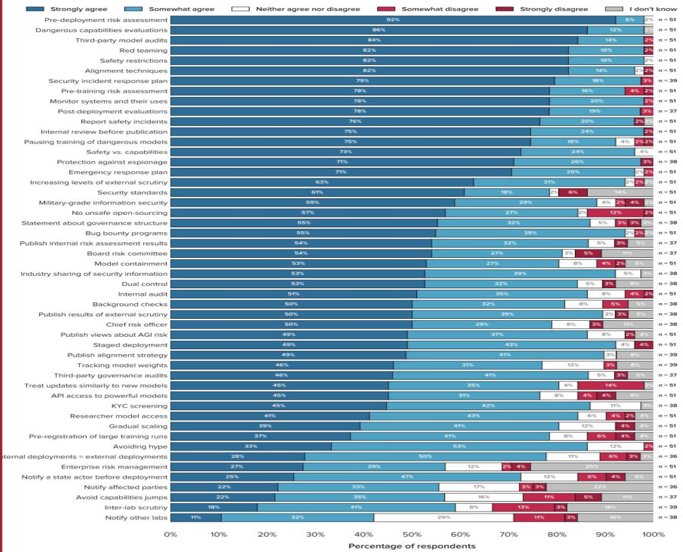

Figure 2: **Percentages of responses for all statements** | The figure shows the percentage of respondents choosing each answer option. At the end of each bar we show the number of people who answered each item. The items are ordered by the total number of respondents that "strongly" agreed. The full statements can be found in Appendix B.

✍🏽 [Schuett, J., Dreksler, N., Anderljung, M., McCaffary, D., Heim, L., Bluemke, E., & Garfinkel, B. (2023). Towards best practices in AGI safety and governance: A survey of expert opinion. arXiv. https://arxiv.org/abs/2305.07153](https://arxiv.org/abs/2305.07153)

Effective Mitigations for Systemic Risks from General-Purpose AI

1. Survey of 76 experts across AI safety, infrastructure, democratic processes, CBRN risks, and bias evaluated 27 mitigations for systemic AI risks.
2. Three measures received highest ratings: safety incident reports/information sharing, third-party pre-deployment audits, and pre-deployment risk assessments with >60% expert agreement.
3. Experts identified external scrutiny, proactive evaluation, and transparency as key principles for effective systemic risk mitigation across all risk domains.
4. Study provides policy recommendations for implementing top-rated measures to inform regulatory frameworks and industry practices for general-purpose AI.

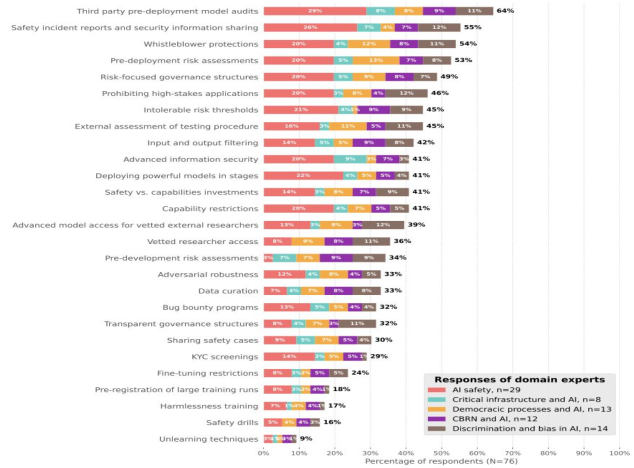

Figure 5: Most selected measures for reducing systemic risk effectively. Experts were asked what combination of ten measures they believed would be most effective at reducing the systemic risks from general-purpose AI. The Figure shows the percentage of the total experts (n=76) who included each measure in their ten measures (indicated at the end of each bar). The colour of the bar represents the expert group, though note that there were different numbers of experts in each group so that the label of each bar segment does not represent the percentage of the expert group who chose the measure.

✍🏽 Uuk, R., Lam, H., Simeon, V., & O'Brien, N. (2024). Effective mitigations for systemic risks from general-purpose AI. arXiv. https://doi.org/10.48550/arXiv.2412.02145

The Unified Control Framework: Establishing a Common Foundation for Enterprise AI Governance, Risk Management and Regulatory Compliance

1. Current AI governance suffers from fragmentation across risk frameworks, varying jurisdictions, and high-level standards lacking implementation guidance, creating false innovation-responsibility trade-offs
2. Unified Control Framework (UCF) integrates risk management and compliance through comprehensive risk taxonomy, structured policy requirements, and 42 unified controls
3. Framework addresses multiple risk scenarios and compliance requirements simultaneously, validated through mapping to Colorado AI Act regulations
4. UCF reduces governance duplication, ensures comprehensive coverage, and enables automation while maintaining innovation speed and responsible AI practices

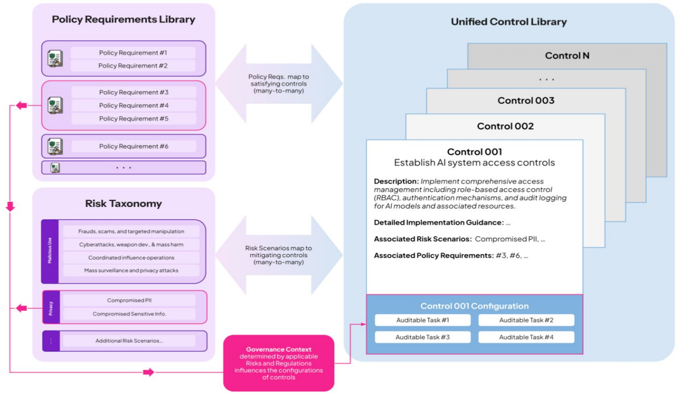

Figure 1: The unified control framework establishes bidirectional mappings between policy requirements, risk scenarios, and controls. (Left) The policy requirements Library contains structured requirements derived from regulations and organizational policies, while the risk taxonomy organizes scenarios into hierarchical categories (e.g., Malicious Use, Privacy). (Right) The unified control library houses implementable controls with detailed specifications, each potentially satisfying multiple policy requirements and mitigating multiple risks. (Center) The governance context determines how these controls should be configured based on the specific combination of applicable regulations and identified risks.

✍🏽 Eisenberg, C., Seaver, A., & Rubin, J. (2025). The unified control framework: Establishing a common foundation for enterprise AI governance, risk management and regulatory compliance. arXiv. https://arxiv.org/abs/2503.05937

Risk sources and risk management measures in support of standards for general-purpose AI systems

1. Comprehensive catalog documents risk sources and management measures for general-purpose AI systems across technical, operational, and societal domains
2. Covers risks throughout model development, training, and deployment stages with descriptive examples and established/experimental mitigation method
3. First extensive documentation providing neutral, self-contained framework independent of existing regulatory approaches for GPAI systems
4. Released under public domain license to support AI providers, researchers, policymakers, and regulators in identifying and mitigating systemic AI risks

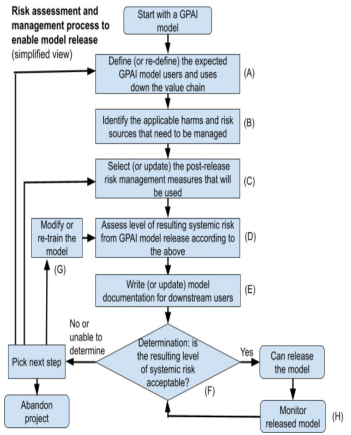

Figure 1: Iterative safety engineering process that can be used by GPAI model providers. This paper has content that is specifically designed to support the engineering steps (B), (C), and (G). Note that we presented a simple flow of step (H), but it can vary depending on the monitoring strategy.

✍🏽 Gipiškis, D., et al. (2024). Risk sources and risk management measures in support of standards for general-purpose AI systems. arXiv. https://arxiv.org/abs/2410.23472

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Pitfalls of Evidence-Based AI Policy

1. Nations face technical uncertainty in AI governance amid "evidence-based policy" debates, but high evidentiary standards risk neglecting risks as seen in tobacco/fossil fuel regulatory delays
2. Study proposes 15 evidence-seeking regulatory goals across institutions, documentation, accountability, and risk-mitigation to facilitate identifying and studying AI risks
3. Key measures include AI governance institutes, model registration, third-party assessments, post-deployment monitoring, security documentation, and whistleblower protections
4. Brazil, Canada, China, EU, South Korea, UK, and USA have substantial opportunities to adopt these process-based, risk-agnostic policies that current evidence gaps make more necessary, not less

	Brazil	Canada	China	EU	Korea	UK	USA
1. AI governance institutes	O*	✔	O*	✔	✔	✔	✔
2. Model registration	✗	✗	✔	✔	✔	✗	O*
3. Model specification and basic info	O*	✗	O	✔	O	✗	✗
4. Internal risk assessments	✔*	✗	O	✔	✔	✗	✗
5. Independent 3rd-party risk assessments	✗	✗	O	O	O	✗	✗
6. Plans to minimize risks to society	O*	✗	O	✔	O	✗	✗
7. Post-deployment monitoring reports	✗	✗	✗	✔	✗	✗	✗
8. Security measures	✗	✗	O	✔	✗	✗	✗
9. Compute usage	✗	✗	✗	O	✗	✗	O*
10. Shutdown procedures	✔*	✗	✗	O	✗	✗	✗
11. Documentation availability	✗	✗	O	O	O	✗	✗
12. Documentation comparison in court	✗	✗	✗	✗	✗	✗	✗
13. Labeling AI-generated content	✗	✗	✔	O	O	✗	✗
14. Whistleblower protections	✗	✗	✔	✔	✗	✗	✗
15. Incident reporting	✔*	✗	✗	✔	✗	✗	✗

Table 1: ✔= Yes, O= Partial, ✗= No, ∗ = proposed but not enacted. There is significant room for progress across the world on passing evidence-seeking AI policy measures. See details on each row above. Note that this table represents a snapshot in time (February 2025). In the USA, we omit the two bills (USA, 2024b;c) discussed above that were proposed last congress but have not been scheduled for re-introduction this congress.

✍🏽 Casper, S., et al. (2025). Pitfalls of evidence-based AI policy. arXiv. https://arxiv.org/abs/2502.09618

A Frontier AI Risk Management Framework: Bridging the Gap Between Current AI Practices and Established Risk Management

1. Comprehensive risk management framework for frontier AI development integrates established principles from high-risk industries with AI-specific practices
2. Four-component framework includes risk identification (literature review, red-teaming, modeling), quantitative analysis with defined thresholds, treatment through mitigation measures, and governance structures
3. Draws from aviation and nuclear power best practices while addressing AI's unique challenges throughout system lifecycle from planning to deployment
4. Emphasizes conducting risk management prior to final training runs to minimize implementation burden and provide actionable guidelines for AI developers

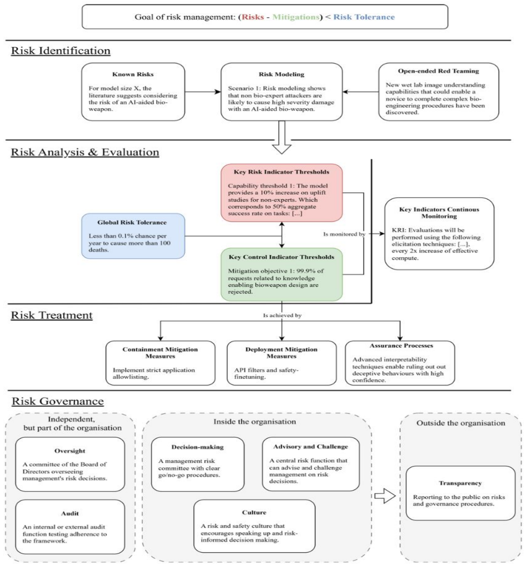

Figure 2: Our complete risk management framework along with examples illustrating each component.

✍🏽 Campos, M., Stewart, A., & Zhang, R. (2025). A frontier AI risk management framework: Bridging the gap between current AI practices and established risk management. arXiv.https://arxiv.org/pdf/2502.06656

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International AI Safety Report 2025

1. First comprehensive AI risk synthesis by 100 experts from 30+ countries, chaired by Turing Award winner Yoshua Bengio
2. Addresses three questions: AI capabilities, associated risks, and available mitigation techniques for general-purpose AI systems
3. Provides scientific evidence without policy recommendations to support informed international policymaking and discussion
4. Independent academic work presented at Paris AI Action Summit (February 2025) following Seoul interim report (May 2024)

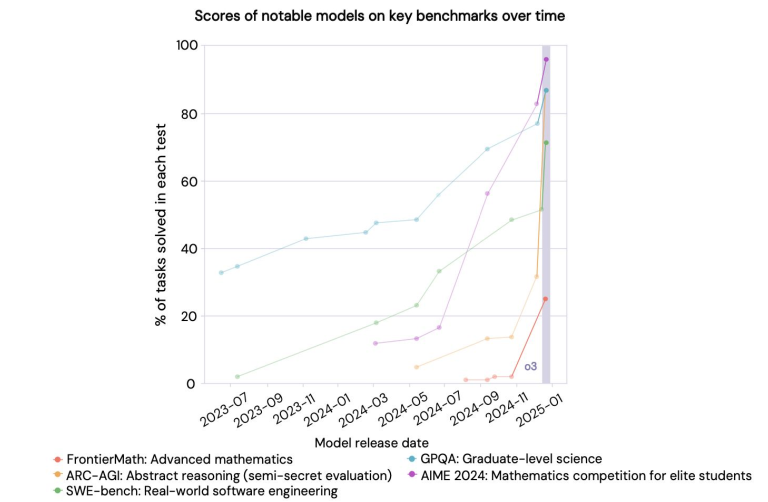

Figure 0.1: Scores of notable general-purpose AI models on key benchmarks from June 2023 to December 2024. o3 showed significantly improved performance compared to the previous state of the art (shaded region). These benchmarks are some of the field's most challenging tests of programming, abstract reasoning, and scientific reasoning. For the unreleased o3, the announcement date is shown; for the other models, the release date is shown. Some of the more recent AI models, including o3, benefited from improved scaffolding and more computation at test-time. Sources: Anthropic, 2024; Chollet, 2024; Chollet et al., 2025; Epoch AI, 2024; Glazer et al. 2024; OpenAI, 2024a; OpenAI, 2024b; Jimenez et al., 2024; Jimenez et al., 2025.

✍🏽 Bengio, Y., et al. (2025). International AI safety report. UK AI Safety Institute. https://www.gov.uk/government/publications/international-ai-safety-report-2025

AI Risk Management Standards Profile for GPAIS

1. Risk-management practices for general-purpose AI systems with beneficial capabilities but potential adverse consequences.
2. Targets large-scale GPAIS developers and downstream application builders using GPAIS platforms.
3. Adapts NIST AI Risk Management Framework and ISO/IEC 23894 for GPAIS-specific issues.
4. Provides practices for identifying, analyzing, and mitigating risks unique to GPAIS development.

Table A4A-2: Summary of Profile Guidance Testing Ratings

High-Priority AI RMF Subcategories	GPT-4	Claude 2	PaLM 2	Llama 2
Govern				
Govern 2.1: Risk assessment and risk management	High	High	High	Medium
Govern 4.2: Report on AI system risk factors	Medium	High	Medium	Medium
Map				
Map 1.1: Identify potential uses/misuses and other impacts	Medium	High	Medium	High
Map 1.5: Set risk-tolerance thresholds for unacceptable risks	Unclear	Unclear	Unclear	Unclear
Map 5.1: Estimate likelihood and magnitude of impacts	Medium	Unclear	Unclear	Unclear
Measure				
Measure 1.1: Tracking important risks: Metrics and red teaming	High	High	Medium	High
Measure 3.2: Tracking elusive risks: Qualitative mechanisms	Medium	Medium	Unclear	High
Manage				
Manage 1.1: Go/no-go decisions	Unclear	Unclear	Unclear	Unclear
Manage 1.3: High-priority risk controls	Medium	Medium	Medium	Medium
Manage 2.3: Unforeseen risk controls	Medium	Medium	Unclear	Low
Manage 2.4: System update and emergency shutdown controls	Unclear	Unclear	Unclear	Low

✍🏽 Barrett, A. M., Newman, J., Nonnecke, B., Hendrycks, D., Murphy, E. R., & Jackson, K. (2024). AI risk management standards profile for GPAIS. UC Berkeley Center for Long-Term Cybersecurity. https://cltc.berkeley.edu/wp-content/uploads/2023/11/Berkeley-GPAIS-Foundation-Model-Risk-Management-Standards-Profile-v1.0.pdf

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FLI AI Safety Index 2024

1. Independent panel of seven AI experts evaluated safety practices of six leading AI companies across critical domains, finding large risk management disparities
2. All flagship models vulnerable to adversarial attacks despite some companies establishing safety frameworks while others lack basic precautions
3. Current strategies deemed inadequate for ensuring AGI systems remain safe and under human control despite companies' explicit AGI development ambitions
4. Companies unable to resist profit-driven safety shortcuts without independent oversight, with experts calling for third-party validation across all firms

Firm	Overall Grade	Score	Risk Assessment	Current Harms	Safety Frameworks	Existential Safety Strategy	Governance & Accountability	Transparency & Communication
Anthropic	C	2.13	C+	B-	D+	D+	C+	D+
Google DeepMind	D+	1.55	C	C+	D-	D	D+	D
OpenAI	D+	1.32	C	D+	D-	D-	D+	D-
Zhipu AI	D	1.11	D+	D+	F	F	D	C
x.AI	D-	0.75	F	D	F	F	F	C
Meta	F	0.65	D+	D	F	F	D-	F

Grading: Uses the US GPA system for grade boundaries: A+, A, A-, B+, [...], F letter values corresponding to numerical values 4.3, 4.0, 3.7, 3.3, [...], 0.

✍🏽 Future of Life Institute. (2024). FLI AI safety index 2024. Future of Life Institute. https://futureoflife.org/document/fli-ai-safety-index-2024/

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EU AI Act: General Purpose AI Code of Practice (Draft 3)

1. EU Code of Practice for general-purpose AI models establishing transparency, copyright compliance, and safety frameworks to demonstrate AI Act compliance
2. All providers must document models, provide downstream information, and implement copyright policies respecting EU intellectual property rights
3. Systemic risk models require comprehensive safety frameworks including risk identification, analysis, mitigation measures, and security protections throughout model lifecycle
4. Mandatory reporting to AI Office through Safety and Security Model Reports, serious incident documentation, and clear organizational responsibility allocation for systemic risk management

✍🏽 EU AI Office. (2025). EU AI Act: General purpose AI code of practice (Draft 3). European Commission. https://code-of-practice.ai/

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Emerging Processes for Frontier AI Safety

1. UK-hosted AI Safety Summit document outlines emerging processes for managing frontier AI risks, including misuse and loss of control.
2. First comprehensive overview of frontier AI safety practices organized thematically to guide policy understanding and comparison.
3. Encourages transparency through published safety policies to build public trust and enable sharing of best practices.
4. Supports proportionate, capability-based approaches that balance innovation with appropriate safety guardrails.

✍🏽 UK Department for Science, Innovation and Technology. (2023). Emerging processes for frontier AI safety. https://www.gov.uk/government/publications/emerging-processes-for-frontier-ai-safety

Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1)

1. NIST cross-sectoral profile implementing AI Risk Management Framework for Generative AI, created pursuant to President Biden's Executive Order on AI safety
2. Defines GAI-specific risks and provides suggested actions to govern, map, measure, and manage risks across AI lifecycle stages and sectors
3. Developed through NIST's Generative AI Public Working Group focusing on four primary considerations: governance, content provenance, pre-deployment testing, and incident disclosure
4. Serves as voluntary companion resource to help organizations incorporate trustworthiness into GAI design, development, use, and evaluation with future revisions planned as technology evolves

✍🏽 National Institute of Standards and Technology. (2024). Artificial intelligence risk management framework: Generative artificial intelligence profile (NIST AI 600-1). NIST. https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

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California State Senate Bill 1047

1. California's SB 1047 requires frontier AI developers to implement shutdown capabilities, safety protocols, and annual third-party audits with Attorney General oversight
2. Prohibits deployment of models with unreasonable critical harm risk; mandates incident reporting within 72 hours and whistleblower protections for employees
3. Creates Board of Frontier Models to update regulations and requires computing cluster operators to assess customer training intentions
4. Establishes CalCompute consortium for public cloud computing framework supporting safe, ethical AI research with civil penalties up to 30% of training costs for violations

✍🏽 Wiener, S. (2024). California State Senate Bill 1047: Safe and Secure Innovation for Frontier Artificial Intelligence Models Act. California State Legislature. https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240SB1047

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How to engage

- Read our evidence scan
- Visit our website
- Offer feedback

Changelog

2025-08-01 - Initial publication of 13 frameworks from evidence scan; Sophia Lloyd George